IQE.L - IQE plc | Full investment artifact

Full investment research | Model assumptions and source evidence

Evidence and full analytical rescore: 19 October 2026 | **Capital anchor:** 2 October 2026 | **Operating case retained:** 17 September 2026 | **Framework:** 8 | **Format:** full current-state analysis **Status:** NOT HELD / WATCHLIST in supplied original | **Classification:** BOTTLENECK CANDIDATE - qualified compound-semiconductor epitaxy **Artifact Base:** 42/60 = **70% - Watchlist** | **Protected current Excel Base:** 70%; the older 77 wording is stale after owner override removal **Snapshot:** frozen supplied-workbook price GBp 47; economic share proxy about 1,412 million; implied capitalization about GBP 664 million using that modeled denominator. No current-market refresh or new holding inference is made. **Active ceiling:** Q4 2027, about GBp 47, pricing Q4 2027-Q3 2028. All fifteen ceiling windows are retained and derived below. The target and operating inputs remain unchanged after this review.

The opportunity extends across more optical architectures and a larger qualified customer base. The full model allows substantial InP growth while preserving the now-visible operating loss and depreciation burden at the starting point. Customer funding improves execution prospects; it does not guarantee margins or a public-share price floor.

New October 19 supply-risk review

The supplied P Equity Research expert call alleges that UK IQE InP polycrystal production is largely halted because Japanese electronic-grade red phosphorus is unavailable. The anonymous speaker gives no named plant, halt dates, lost order amount or current stock cover. This is a specific adverse check worth investigating; it does not establish the operating status of every group product line.

Issuer sources distinguish Wafer Technology raw/substrate activity, internal and external customers, and the group's commercial epitaxial wafer business. The catalog describes high-purity InP polycrystal synthesized in an HHP furnace and supplied as source material for LPE/MBE. Current model receipts are allocated to InP optical epitaxy, other photonics and wireless. No published mapping identifies how much modeled epi output depends on the alleged halted raw supply, which customers own the substrates, or which qualified alternatives/buffers exist. The earlier South Wales silicon closure is a different site/product event.

Joint impairment probability 15%. This subjective watch probability describes a current raw-poly interruption causing net group OP loss beyond the existing path. It carries limited weight because the original expert is anonymous and the company-specific receipt/procurement link is unresolved. It does not describe all epitaxy stopping. The September interim's H2 tooling-conversion plan and undated product catalog do not prove October supply availability; official silence is not the basis for zero.

We attempted two bounded cost bridges: a narrow epi-procurement shortage and direct external raw-material sales. The first varies exposed InP receipts from 5-25%, with 10-40% shortfall after an assumed 50% buffer. Half reference COGS is avoidable; fixed manufacturing/depreciation remains, plus 1% qualification cost on the exposed pool. The second tests 5-20% of other-photonics receipts as raw-sales exposure with 60% avoidable variable cost. Neither allocation is a reported revenue split. They are alternative overlapping routes, with no sum into the active model.

The supplier/site/product mapping is the missing variable: selecting a point loss would assign raw-poly activity to epi or other-photonics revenue without evidence. The sensitivity demonstrates the cost mechanism and guides retrieval; it does not establish that linkage. Active revenue, GP, opex, shares and all targets therefore remain unchanged. The existing substrate-supply risk already deducts one point within the maximum five-point total.

expected to see clearer progress. Several domestic companies are also developing diamond wire cutting equipment for InP, including Hunan Yujing, Wuxi Liancheng and Jingsheng Mechanical & Electrical.

Q16. What is the competitive landscape of the domestic InP substrate market, and what are the shipment volumes of the main manufacturers?

In the domestic market, Yunnan Germanium has relatively large shipments, and the gap between Xiandao Optoelectronics and Yunnan Germanium is not large; both companies reach monthly shipments of 20,000–30,000 wafers. Other manufacturers are far behind these two.

Q17. What is the current supply relationship between upstream InP polycrystal suppliers and downstream substrate makers?

There is a strong binding relationship between upstream InP polycrystal suppliers and downstream substrate makers. There are currently three main polycrystal suppliers: the UK's IQE is basically halted because it cannot obtain red phosphorus from Japan; Beijing Mingjia Semiconductor ships 2 tons of InP polycrystal per month, with capacity prioritized for Yunnan Germanium; Shaanxi Yinjie mainly supplies Yunnan Germanium and AXT Crystal. AXT has some polycrystal synthesis capability of its own, but with small capacity. This tight binding makes it hard for new substrate entrants to obtain a stable raw material supply chain, so many companies have had to set up their own polycrystal synthesis business.

Q18. Given the current tight supply, what is the customer qualification process for InP substrates, and what are the yield risks of the shortening qualification cycle?

Customer qualification cycles are indeed continuing to shorten. A few years ago (2024–2025) the cycle was about two years, whereas now, for larger companies such as Jiangsu Huaxin Semiconductor and

Original expert call page14,published19October2026. Anonymous IQE raw-poly assertion with no quantified site/output scope.

IQE watch control	Current treatment
Impairment probability	15% subjective joint outcome
Exact affected plant/halt period	Missing
Raw versus epi receipt mapping	Missing
Qualified material stock and alternative supply	Missing
Group H1 inventory	About GBP 20 million; material/site cover unknown
Active operating increment	0; scope gap after bounded proxy attempt
Analytical rescore	42/60=70, unchanged
Current Excel Base	70 protected

Diagnostic only	Exposed receipt pool	Loss after buffer	Near Q4 2026 target impact
Epi procurement, low	5%	10%	Down less than GBp 1
Epi procurement, middle	10%	20%	Down less than GBp 1
Epi procurement, high	25%	40%	Down less than GBp 1
Direct raw sales alternative	5-20% other photonics	Full exposed receipt loss	Down less than GBp 1

The transient diagnostic periods are Q4 2026/Q1 2027. They do not touch the active Q4 2027 valuation window. Extending a halt into that later window needs new duration/customer evidence. Exact conditional and probability-weighted receipt/cost figures remain in the model appendix.

1. Supercycle, supply chain and durable advantage

IQE supplies qualified epitaxial materials used in optical and RF devices. Customer agreements and financing strengthen its capacity-conversion opportunity. The central question is whether higher photonics loading can absorb manufacturing depreciation and corporate costs while generating cash. Its strategic review has concluded; current valuation rests on operating execution.

DB1 photonics: optical systems → lasers, modulators and detectors → device fabrication on engineered epi layers → qualified epitaxy grown on raw substrates. The modeled vector captures epi receipts. The group also owns substrate and source-material operations, so raw-material, substrate, epi and completed-device economics need separate boundaries. Recipes, uniformity, defect control and customer qualification create switching costs, while competing epitaxy and internal production limit exclusivity. GaAs QDL and InP can serve different or overlapping optical architectures.

Business / curve	Stage	Evidence and next inflection
InP optical epitaxy	Early commercial ramp;10-50% qualitative	Signed supply relationships and conversion of tooling support growth.
Wireless and legacy photonics	Recovery/mature applications	Handset, infrastructure and defense mix drive utilization.
QDL and newer platforms	Pre-ramp 0-10%	Customer sampling must convert into product launches and recurring volume.

Stage percentages are qualitative framework bands relative to a possible mature opportunity, not measured market penetration. A current capacity constraint can coexist with a larger long-term TAM; qualification and economic capture decide how much reaches this company.

2. Forward TAM and the three-phase expansion lens

Phase	Engineering opportunity	Demand-to-company bridge
Phase 1;100% thematic	Existing optical/RF epi demand	More optical links and bandwidth create additional wafer demand. IQE captures epi invoice content, not the value of complete transceivers.
Phase 2;50% thematic	Higher-speed photonics, OCS and broader customers	Tower's agreement supports a qualified route into next-generation platforms. Modeled InP80 M 2027→285 M 2030 requires about 4x revenue. An illustrative 3x sellable wafer-equivalent volume and about 19% higher content/net ASP gives that endpoint; both factors are analyst assumptions.
Phase 3;20% thematic	QDL, microLED, advanced sensing and GaN applications	Physical performance can create new product markets, but qualification, wafer economics and customer demand decide capture. QDL is within other photonics; no full global photonics-TAM overlay is added.

The 100% / 50% / 20% thematic weights guide thesis assessment. They are not blanket discounts to reported sales, contracts or all forecasts. Programme probabilities below govern only separately identified optionality. Current reported sales calibrate the starting point; they do not cap the future market.

3. Current earnings and management evidence

H1 revenue was GBP 65 M: Photonics 39 M and Wireless 26 M. Reported GP/OP were 6 M/-8 M; adjusted GP/OP were 7 M/-4 M. Adjusted EBITDA 6 M less D&A 10 M plus <1 M disposal gain reconciles to adjusted OP. Cash was 42 M; net debt including leases was 12 M.

September 7 interim financial statements and management outlook were reviewed against the May 28 FY2025 release. Full webcast Q&A transcripts were not obtained; this is a filings-and-management-release review. The latest outlook retains FY2026 revenue growth above 30% and low-teens adjusted EBITDA, not operating profit.

Latest interim and annual statements plus management releases were reviewed. Complete webcast Q&A transcripts were not obtained, and this is explicitly not claimed as two full earnings-call reads. Product-level revenue, costs, precise capacity and the complete historical option schedule remain evidence gaps.

4. Local evidence and calibration packet

H1 accounts replace a preliminary revenue-only anchor with disclosed GP, operating loss, D&A and cash. The model starts in H2, so both first rows are forecasts. H2 sales 34 M/37 M and OP-2 M/-1 M reconcile with an approximately 13 M FY adjusted EBITDA case when H2 D&A equals H1. This meets the low-teens EBITDA direction without treating EBITDA as operating profit. The going-concern planning assumptions are distinguished from formal long-term public guidance.

5. Summing thesis score

Category	Points / maximum	Evidence
Core Thesis	14/20	Customer financing and signed supply agreements validate capability, with competing epi suppliers.
Cycle	8/10	Recovery is printing, but adjusted operating losses remain.
TAM	9/10	InP, QDL, GaN power/RF and microLED expand applications without implying full end-market capture.
Catalysts	4/5	InP tooling conversion and customer sampling; prior sale catalyst has ended.
Macro	9/10	AI optics and defence demand support qualified epitaxy.
Alpha	3/5	Margin recovery opportunity, tempered by public awareness of the financing and supply deals.
Risks	-5	Qualification/yield, customer concentration, substrate supply, margin/cash conversion and dilution.

Sum: 42/60 × 100 = 70%, conventionally rounded to 70%. This advisory score remains 70 after a fresh canonical review; current Excel 70 is protected. The separate risk classes below total the displayed deduction; holding status and classification do not waive that deduction.

6. Revenue sources, capture and product margins

Calendar quarters. Keep 20x adjusted operating profit and multiply GBP/share by 100 for GBp. H2 2026 revenue is GBP 34 M/37 M, giving FY2026 GBP 136 M, about 39% growth. H2 OP is -2 M/-1 M, not breakeven. With H2 D&A assumed equal to H1 (10 M), FY adjusted EBITDA is about 13 M, within low teens. Product GP includes manufacturing depreciation; corporate operating costs include relevant administrative amortisation. No D&A add-back enters the valuation. The H2 InP/other-photonics/wireless allocations are analyst proxies, calibrated to those OP assumptions. Annual revenue vectors for 2027-30: InP 80/145/215/285 M, other photonics 42/48/54/60 M, wireless 58/64/71/78 M. Weights 22/24/26/28%. Product GM ramps only as displayed, with InP easing from 38% in 2029 to 35% in 2030. Corporate/operating expense is GBP 22/26/30/34 M annually. These are conditional high-utilization economics, not management guidance. Tower and MACOM sales are within InP; QDL, storage and other optical technologies are within other photonics, with no additional contract layer. The share proxy adds full MACOM conversion and June awards to September 30 non-treasury issued shares. Associated MACOM warrants expire on conversion: do not automatically count both as simultaneous dilution. The complete historical option inventory remains unresolved; the denominator is a specified minimum forward dilution scenario rather than a complete accounting diluted share count.

Vector	Actual calibration / source	Forward demand and capture	Margin derivation / constraint
InP optical epitaxy	12 M Q3 allocation within 21 M modeled photonics; product actual split is not disclosed.	Annual 80/145/215/285 M GBP 2027-30. The 4×2030/2027 bridge requires substantially more qualified output and richer content. At a purely illustrative 1,000 GBP per wafer-equivalent invoice,80 k→285 k units are required; actual wafer diameter, process scope and ASP remain unknown.	Quarterly 24/27/30/33% GM in 2027;36% 2028,38% 2029,35% 2030. Rising loading reduces fixed cost per sale, followed by modest price/mix normalization. At 1,000 GBP invoice and 36% GM,640 GBP productcost is the implied requirement, not disclosed unit cost.
Other photonics	9 M initial allocation; includes IR/VCSEL/storage and QDL.	42/48/54/60 M annual sales. The 14 M USD multi-year storage order is included within this envelope, not counted each year or relabeled InP. QDL sampling supports a prospective mix shift but has no disclosed material revenue schedule.	16% 2027→22% 2029-30 product GM proxy. New products may have lower initial yields; mature applications offset some ramp costs.
Wireless	13 M initial quarter; H1 reported 26 M.	58/64/71/78 M annual sales;1×2027-30 recovery in qualified RF demand and content. No whole GaN power-market capture is assumed.	6% 2027→10% 2029-30 product GM reflects recovery from low loading. This remains the lower-margin business in the model.

Confidence: reported historical totals are HIGH. Source-based forward extrapolations are MEDIUM only where the cited source sizes the relevant period; other product allocations, future capture and margin assumptions are LOW. A source for the historical anchor does not make an unsourced future factor management guidance. The detailed factors below are the single modeled base case, with uncertain assumptions exposed for review.

V1. InP optical epitaxy (within Photonics)

Sources: I1/2/3; InP customer agreements and assumed throughput/content. Revenue = anchor × displayed reference percentage /100; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor	Revenue vs anchor	Revenue	Product GM	GP / contribution
Q3 2026	12	100%	12	17%	2
Q4 2026	12	117%	14	18%	3
Q1 2027	12	147%	18	24%	4
Q2 2027	12	160%	19	27%	5
Q3 2027	12	173%	21	30%	6
Q4 2027	12	187%	22	33%	7

Quarter	Revenue anchor	Revenue vs anchor	Revenue	Product GM	GP / contribution
Q1 2028	12	266%	32	36%	11
Q2 2028	12	290%	35	36%	13
Q3 2028	12	314%	38	36%	14
Q4 2028	12	338%	41	36%	15
Q1 2029	12	394%	47	38%	18
Q2 2029	12	430%	52	38%	20
Q3 2029	12	466%	56	38%	21
Q4 2029	12	502%	60	38%	23
Q1 2030	12	522%	63	35%	22
Q2 2030	12	570%	68	35%	24
Q3 2030	12	618%	74	35%	26
Q4 2030	12	665%	80	35%	28

V2. Other Photonics: IR, VCSEL, QDL and storage

Sources: I1/4/5; IR/VCSEL base plus storage and QDL within the same allocation. Revenue = anchor × displayed reference percentage /100; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor	Revenue vs anchor	Revenue	Product GM	GP / contribution
Q3 2026	9	100%	9	12%	1
Q4 2026	9	106%	10	12%	1
Q1 2027	9	103%	9	16%	1
Q2 2027	9	112%	10	16%	2
Q3 2027	9	121%	11	16%	2
Q4 2027	9	131%	12	16%	2
Q1 2028	9	117%	11	20%	2
Q2 2028	9	128%	12	20%	2
Q3 2028	9	139%	12	20%	2
Q4 2028	9	149%	13	20%	3
Q1 2029	9	132%	12	22%	3

Quarter	Revenue anchor	Revenue vs anchor	Revenue	Product GM	GP / contribution
Q2 2029	9	144%	13	22%	3
Q3 2029	9	156%	14	22%	3
Q4 2029	9	168%	15	22%	3
Q1 2030	9	147%	13	22%	3
Q2 2030	9	160%	14	22%	3
Q3 2030	9	173%	16	22%	3
Q4 2030	9	187%	17	22%	4

V3. Wireless epitaxy

Sources: I1/7; wireless recovery and qualified product demand. Revenue = anchor × displayed reference percentage /100; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor	Revenue vs anchor	Revenue	Product GM	GP / contribution
Q3 2026	13	100%	13	4%	<1
Q4 2026	13	104%	14	4%	<1
Q1 2027	13	98%	13	6%	<1
Q2 2027	13	107%	14	6%	<1
Q3 2027	13	116%	15	6%	<1
Q4 2027	13	125%	16	6%	<1
Q1 2028	13	108%	14	8%	1
Q2 2028	13	118%	15	8%	1
Q3 2028	13	128%	17	8%	1
Q4 2028	13	138%	18	8%	1
Q1 2029	13	120%	16	10%	2
Q2 2029	13	131%	17	10%	2
Q3 2029	13	142%	18	10%	2
Q4 2029	13	153%	20	10%	2
Q1 2030	13	132%	17	10%	2
Q2 2030	13	144%	19	10%	2

Quarter	Revenue anchor	Revenue vs anchor	Revenue	Product GM	GP / contribution
Q3 2030	13	156%	20	10%	2
Q4 2030	13	168%	22	10%	2

7. Operating-cost bridge and quarterly operating model

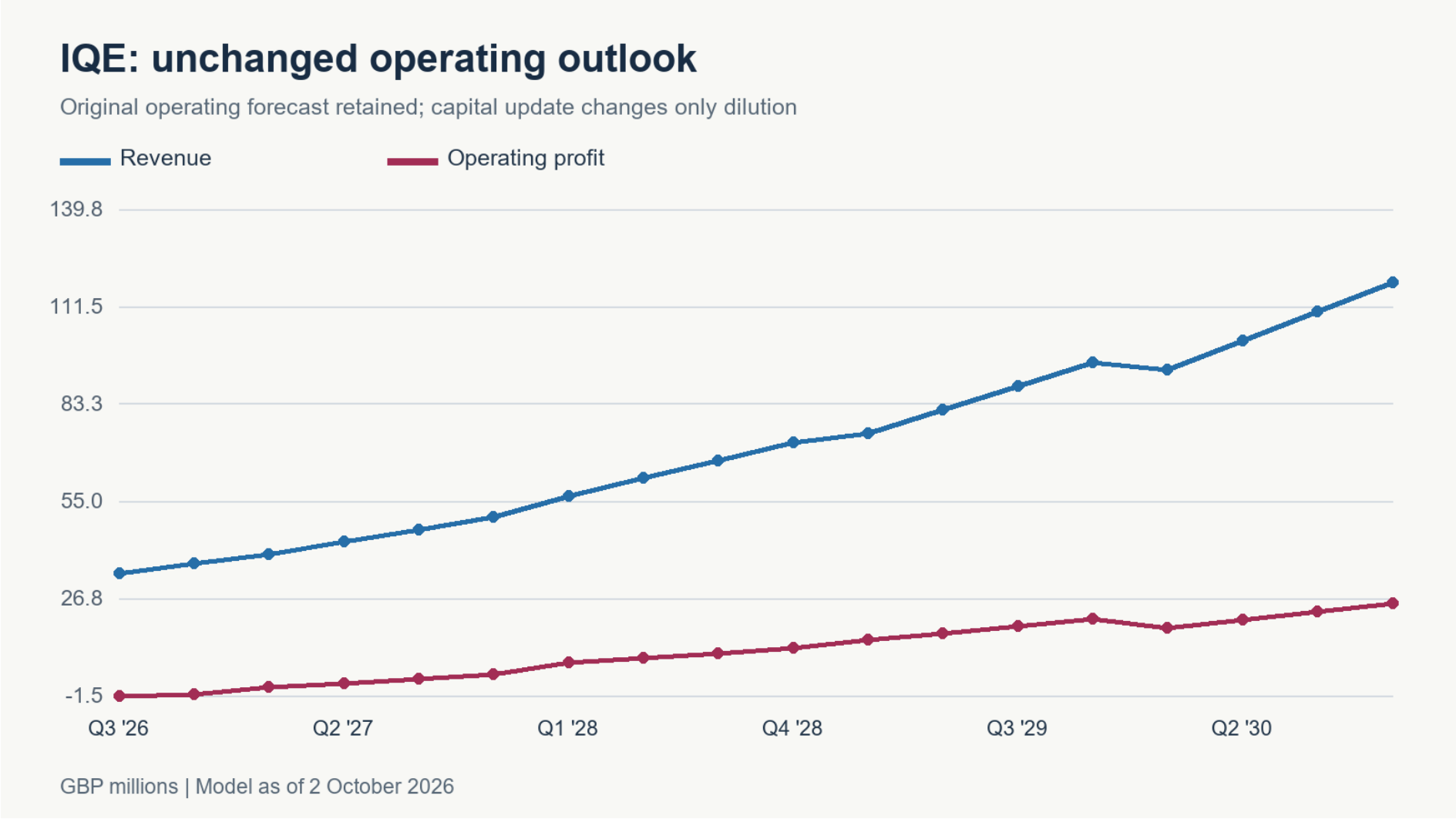
Explicit annual expenses 22/26/30/34 M GBP 2027-30 rise despite a declining ratio. Product margins include manufacturing depreciation and relevant amortization remains in operating costs. H1 adjusted EBITDA6.036 M-D&A9.789 M+disposal <1 M=-4 M adjusted OP. Adjusted OP itself excludes certain charges that may recur; it is not free cash flow. Future reactor conversion and customer qualification require investment in addition to the accounting profit bridge.

Profit basis: Adjusted operating profit after depreciation and amortisation. Amounts below are GBP millions.

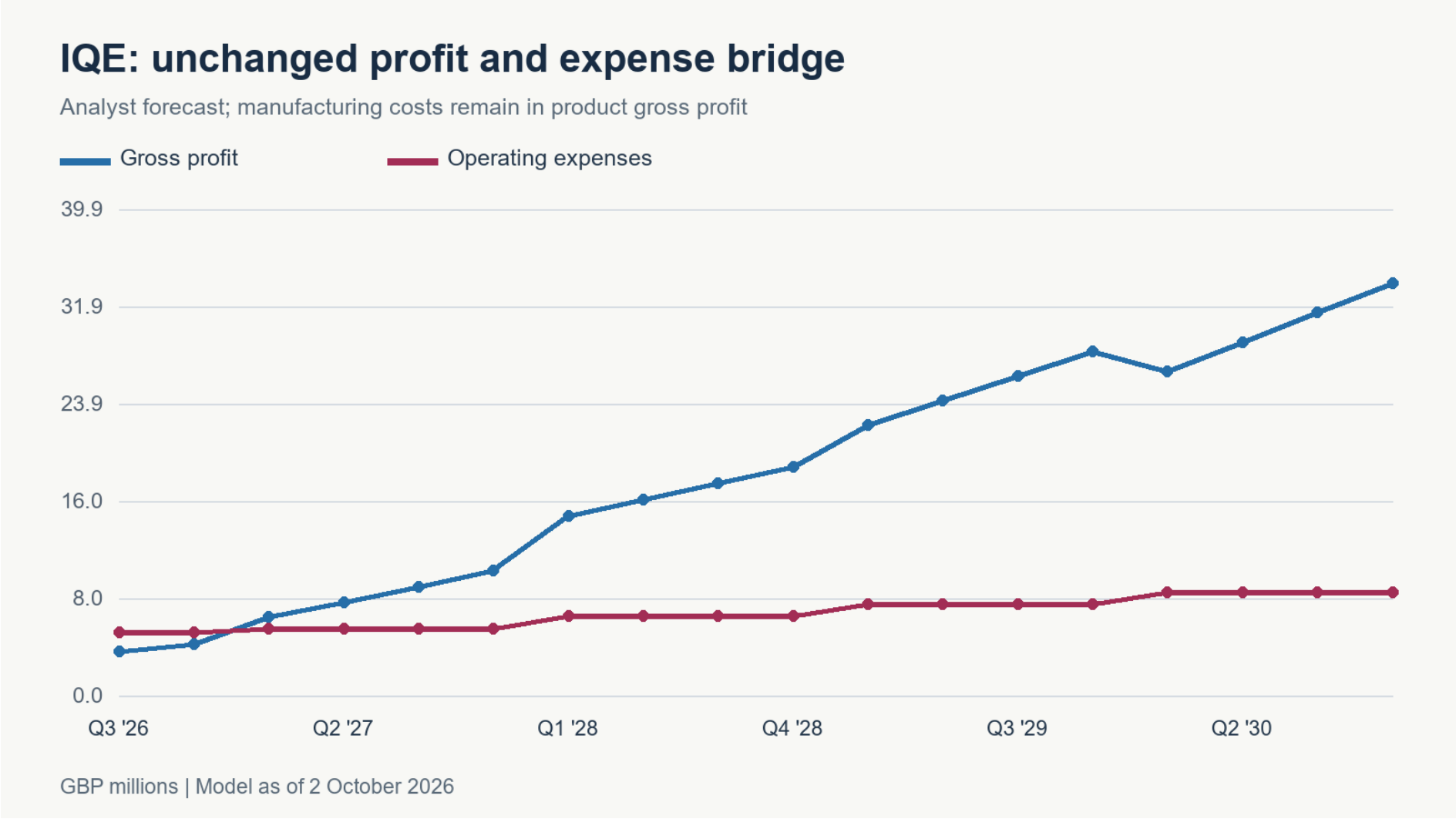
Quarter	Status	Revenue	GP / contribution	Opex %	Opex	OP margin	Operating profit	Main driver	Confidence
Q3 2026	Model	34	4	15%	5	-4%	-2 (neg)	H2 InP ramp with operating loss	LOW model
Q4 2026	Model	37	4	14%	5	-3%	-1 (neg)	Higher sales; D&A still material	LOW model
Q1 2027	Model	40	6	14%	6	2%	<1	2027 qualified volume	LOW model
Q2 2027	Model	43	8	13%	6	5%	2	InP loading improves	LOW model
Q3 2027	Model	47	9	12%	6	7%	3	Higher InP product margin	LOW model
Q4 2027	Model	50	10	11%	6	9%	5	2027 exit production	LOW model
Q1 2028	Model	57	15	11%	6	15%	8	2028 capacity conversion	LOW model
Q2 2028	Model	62	16	11%	6	16%	10	More qualified epi output	LOW model
Q3 2028	Model	67	17	10%	6	16%	11	Customer adoption	LOW model
Q4 2028	Model	72	19	9%	6	17%	12	2028 mix and throughput	LOW model
Q1 2029	Model	75	22	10%	8	20%	15	2029 higher optical content	LOW model

Quarter	Status	Revenue	GP / contribution	Opex %	Opex	OP margin	Operating profit	Main driver	Confidence
Q2 2029	Model	82	24	9%	8	20%	17	Strong utilization case	LOW model
Q3 2029	Model	88	26	8%	8	21%	19	Peak modeled product margin	LOW model
Q4 2029	Model	95	28	8%	8	22%	21	2029 production scale	LOW model
Q1 2030	Model	93	27	9%	8	19%	18	2030 price/mix normalization	LOW model
Q2 2030	Model	102	29	8%	8	20%	20	Continued volume growth	LOW model
Q3 2030	Model	110	31	8%	8	21%	23	Broader photonics demand	LOW model
Q4 2030	Model	118	34	7%	8	21%	25	2030 steady conversion case	LOW model

OP = sum of product GP/contribution - explicit operating costs. OP margin = OP ÷ revenue. Displayed numbers are rounded for reading; the workbook calculates from unrounded inputs. Guidance assigned to a future quarter retains Model status because guidance is not an actual result.



IQE: unchanged operating outlook. GBP millions. Original operating forecast retained; capital update changes only dilution



IQE: unchanged profit and expense bridge. GBP millions. Analyst forecast; manufacturing costs remain in product gross profit

8. Multiple, shares and recognition convention

20×forward adjusted OP reflects qualified compound-semiconductor recipes, optical demand and customer commitments. It is conditional on operating conversion after depreciation. Competitors and capital intensity prevent importing finished-device or memory-vendor margins wholesale. There is no takeover premium or implied strategic-investor price floor.

September 30 issued 1,331,131,442 ordinary shares less 4,115 treasury shares, plus about 75,757,576 MACOM conversion shares and 5,300,788 June employee awards gives the retained economic proxy of about 1,412 million shares. The MACOM GBP 15 million conversion uses approximately 20 p per share; the exact contractual conversion term remains unrounded in the calculation file. Conversion-linked warrants expire on conversion and are not counted twice. This proxy is a specified minimum forward dilution scenario; the full option inventory remains unresolved. H1 cash about GBP 42 million and net debt including leases about GBP 12 million remain historical anchors. [I1/I2/I8]

The valuation denominator is the ceiling quarter's diluted-share proxy. It is not a historical EPS weighted-average assumption disguised as a fully known future capitalization. Unit factor: 100; this converts the model units to GBp per share. No cash balance, enterprise-value adjustment or after-tax earnings substitution is added to the operating-profit ceiling.

Actual capital update and unchanged forward allowances

The September 30 RNS reports 1,331,131,442 ordinary shares of 1 p each, including 4,115 treasury shares, leaving 1,331,127,327 voting rights. The reported issue count is not a weighted-average diluted EPS denominator. The RNS does not specify which instrument caused the 36,293-share increase; this update changes only actual issued capital and retains every existing forward allowance. The unchanged June-award allowance is not recast as newly issued equity. [I8]

Retained capital control	Current amount
September issued ordinary shares	1,331,131,442
Treasury shares	4,115
MACOM conversion allowance, rounded shares	75,757,576
June award allowance, shares	5,300,788
Economic forward share proxy, millions	1,412
Active Q4 2027 F4Q OP, GBP millions	33
Active Q4 2027 ceiling, GBp	47

The October 2 capital-only update added 36,293 issued shares and reduced its target by less than 1%. That historical revision is not repeated now. Exact current capital terms, decimal GBP conversion and historical before/after targets remain in the linked model.

9. Complete rolling four-quarter ceiling derivation



IQE: rolling four-quarter price-target path. GBp per share. Lines nearly overlap; exact capital-only changes are tabulated below

Ceiling quarter	Exact earnings window	Forward 4Q OP	Multiple	Diluted shares M	Ceiling GBp	Key inflection
Q3 2026	Q3 2026 to Q2 2027	<1	20x	1,412	<1	H2 InP ramp with operating loss
Q4 2026	Q4 2026 to Q3 2027	5	20x	1,412	8	Higher sales; D&A still material

Ceiling quarter	Exact earnings window	Forward 4Q OP	Multiple	Diluted shares M	Ceiling GBp	Key inflection
Q1 2027	Q1 2027 to Q4 2027	11	20x	1,412	16	2027 qualified volume
Q2 2027	Q2 2027 to Q1 2028	18	20x	1,412	26	InP loading improves
Q3 2027	Q3 2027 to Q2 2028	26	20x	1,412	37	Higher InP product margin
Q4 2027	Q4 2027 to Q3 2028	33	20x	1,412	47	2027 exit production
Q1 2028	Q1 2028 to Q4 2028	41	20x	1,412	58	2028 capacity conversion
Q2 2028	Q2 2028 to Q1 2029	47	20x	1,412	67	More qualified epi output
Q3 2028	Q3 2028 to Q2 2029	54	20x	1,412	77	Customer adoption
Q4 2028	Q4 2028 to Q3 2029	62	20x	1,412	88	2028 mix and throughput
Q1 2029	Q1 2029 to Q4 2029	71	20x	1,412	100	2029 higher optical content
Q2 2029	Q2 2029 to Q1 2030	74	20x	1,412	105	Strong utilization case
Q3 2029	Q3 2029 to Q2 2030	78	20x	1,412	110	Peak modeled product margin
Q4 2029	Q4 2029 to Q3 2030	82	20x	1,412	116	2029 production scale
Q1 2030	Q1 2030 to Q4 2030	87	20x	1,412	123	2030 price/mix normalization

Each row sums its currentquarter and nextthreequarter OP, then applies 20x and 100 p/GBP, divided by the same economic forward share proxy. For active Q4 2027 the fourquarter OP rounds to GBP 33 million; using unrounded OP and shares gives about GBp 47. Exact eighteen operating rows and fifteen target derivations are retained in the model. The final three quarters lack complete future windows and have no target. All operating, cost, share and multiple inputs remain unchanged in this review.

10. Annual trajectory cross-check

Calendar year	Revenue	GP / contribution	Opex	OP	OP margin
2027	180	33	22	11	6%
2028	257	67	26	41	16%
2029	340	101	30	71	21%
2030	423	121	34	87	21%

These are calendar-year sums of the quarterly path, not extra revenue and not the live valuation method. For non-December fiscal reporters, fiscal guidance and these calendar totals must be aligned using the period mapping above; a four-quarter run rate is not automatically the same fiscal year. Outer-year outputs are conditional TAM/capture trajectories, not verified management sales targets.

Annual company sales 180 M/257 M/340 M/423 M GBP 2027-30 growth is 33% over modeled 2026 sales 136 M, consistent with rather than guaranteed by the internal planning assumptions. InP becomes 67% of 2030 sales, so the case depends on mix change and productive capacity, not uniform growth across all businesses. Customer agreements lack enough public quantities to independently validate 285 M InP sales.

11. Dilution sensitivity - same operating case

Denominator sensitivity	Q4 2027 shares M	Q4 2027 ceiling GBp	Effect on ceiling
-5% shares	1,342	50	+5%
+0% shares	1,412	47	+0%
+5% shares	1,483	45	-5%
+10% shares	1,553	43	-9%

This isolates denominator uncertainty; it is not a second operating case or permission to remove known dilution. The updated share path above remains the active convention. The comparison now uses the current Q4 2027 horizon; operating assumptions are unchanged.

12. Programme and probability ledger

Programme / signal	Source and probability	Amount / unit base	Treatment in this model	Escalation / recognition
Tower supply	Signed customer agreement;85% programme confidence	Minimum commitments with undisclosed amounts	Inside InP; no additive contract windfall	Paid qualified wafer shipments and disclosed quantities calibrate capture.
MACOM capacity	Customer financing;85% deployment confidence	Known funding, unknown detailed revenue schedule	Inside InP and share bridge	Installed/qualified tooling and net sales determine conversion.
Storage order	Announced multi-year production order;85% execution confidence	14 M USD total over multiple years	Inside other photonics, no annual repetition	Delivery and recognition schedule replaces the allocation.
QDL sampling	Named customer purchase agreement;60% material-scale confidence	No public mass-production value	Supports other-photonics capture; zero separate optionality amount	Customer launch and recurring orders upgrade the financial case.
Outer InP ramp	Analyst demand/capture scenario;50% confidence	3×wafer-equivalent volume×1 content	Embedded long-range base path, not a second haircut	Disclosed reactor capacity, yield and ASP must support output.

Probabilities are analyst judgments under the framework signal hierarchy. Embedded revenue is not multiplied or added a second time. Where no defensible amount or unit base exists, the opportunity remains explicitly unquantified; confidence alone cannot create a revenue number. New evidence can enlarge the TAM, improve capture or move recognition earlier, but each requires an identifiable model input.

13. Scored risk classes

Risk class	Scored deduction	Mechanism and consequence
Qualification and yield	-1	A reactor or sample is not qualified sellable production; utilization losses consume cash.
Customer concentration	-1	Funding and purchase commitments can increase customer bargaining power.
Substrate supply	-1	InP input availability, quality and restrictions can limit epi output despite demand.
Margin and cash conversion	-1	Manufacturing depreciation, leases and recurring adjustments limit economic profit.
Dilution	-1	Convertible shares and employee awards reduce common claims; the option inventory is incomplete.

14. Dated catalysts, counters and escalation

1. **2026-12-31 - FY26 production** Test H2 sales 71 M and continuing operating losses against actual shipments and D&A.
2. **2027-03-31 monitoring date - Annual results** Reconcile low-teens EBITDA to OP and cash; refresh all 18 operating quarters.
3. **2027-06-30 - InP conversion** Verify qualified output and actual photonics mix, not installed reactors alone.
4. **2027-12-31 - Customer scale** Check 180 M company sales and rising product GP against spending.
5. **2028-12-31 - High-utilization economics** 36% InP GM requires actual cost absorption, not a shortage narrative alone.
6. **Every capital update - Conversion and awards** Rebuild complete dilution if new instruments, awards or conversion-price changes appear.

15. Cross-book monitoring

The Tower agreement is a verified supplier/customer link. IQE epitaxy is an input to the downstream foundry platform, so the two companies' revenue does not represent independent end-customer capex. InP and GaAs QDL are different technology routes; no assumption that each captures all optical lanes is made. CRDO/MRVL optical adoption expands potential device demand; IQE's wafer revenue still requires a specific qualified process/customer route. Tower's larger SiPho outlook cannot be multiplied by an arbitrary IQE take rate without specifying the InP content, architecture and sourcing share.

16. Source register and model limits

I1 - H1 2026 results and notes (2026-09-07). Notes 7-8 reconcile adjusted GP, EBITDA and OP. The going-concern model assumes at least 30% revenue growth in 2026 and 2027; that internal planning assumption is not a separate long-term public earnings target.

I2 - Strategic review conclusion (2026-04-27). The review concluded with financing. The prior artifact's pending takeover/Taiwan-sale catalyst is removed. Customer investment is not a contractual floor for public shares.

I3 - Tower supply agreement (2026-06-15). Minimum purchases/volumes and reciprocal supply commitments were announced, without disclosed revenue amounts. The agreement also settles the prior IP dispute.

I4 - Production order (2026-07-14). The $14 M order is multi-year and supports storage applications. It is not $14 M annual InP revenue.

I5 - Quintessent QDL agreement (2026-09-03). Six-inch GaAs QDL epitaxy is progressing to customer sampling; the release does not quantify a production revenue ramp.

I6 - August voting rights (2026-09-01). Historical August baseline: issued 1,331 M less <1 M treasury. Replaced in the active bridge by September 30 count I8; MACOM and award allowances retained.

I7 - FY2025 release (2026-05-28). FY2025 revenue GBP 97 M anchors the FY2026 growth comparison.

I8 - September total voting rights, RNS0275X (2026-09-30). Full original story linked through the issuer regulatory-news page:1,331,131,442 ordinary 1 p shares,4,115 treasury and 1,331,127,327 total voting rights as at September 30. Actual issued increment 36,293 from August. No new financing or employee-award forecast is inferred from that count.

No complete two-webcast Q&A review. Product revenue splits, margins, quarterly timing and outer-year expansion are LOW-confidence assumptions. The disclosed multi-year agreements lack a complete revenue schedule. Full future option dilution is not reconstructed. Non-GAAP OP is not cash flow and the framework multiple does not separately adjust for leases or cash.

Workbook: v 3_2_master_portfolio.xlsx, model tab IQE_L_Q_Model. It owns portfolio ranking, prices and return calculations; those are intentionally outside this artifact. **Current artifact path:** G:\My Drive\1 amInvesting\Artifacts\IQEL_IQE_DeepDive.md. **Original snapshot:** versions/originals/IQEL_IQE_DeepDive.md. Superseded history is retained separately in VERSION_LOG, not in this current-state model.

17. New source register and falsification tests

- P Equity Research,19 October 2026, Expert Call|InPSubstrate Market Outlook. Original sixteen-page owner-supplied PDF retained. Page 13 introduces the UKpolycrystal interruption; page 14 namesIQE. Expert identity, recording, exact plant, halt dates and affected quantities are missing. The source's other substrate forecasts do not establish IQE epi revenue.
- IQE substrate operations, undated current catalog. Internal/external substrate customers and UK/US activity; product availability is not certified by a web listing.
- IQE source materials, undated current catalog. HHP InP polycrystal, LPE/MBE feedstock and source material scope.
- IQE InP epi products, undated current catalog. Product scope provides an entity/product cross-check.
- IQE September 7 interim, H2 tooling conversion and financial controls. This older statement does not resolve an Octoberraw-poly halt.

Priority follow-up is the named operating site/product, current production status, external raw-sales amount, and relevant epi customer procurement/stock/alternatives. If a current stoppage causes lost accepted epi shipments or lost external raw invoices, update only that mapped cohort and period with retained fixed costs and incremental qualification charges. Remove the watch if the alleged interruption is stale, belongs to another product/entity or was fully buffered. A complete Q&A, product cost split and historical option inventory remain outstanding as in the full retained research.

Current control: 20x adjusted OP afterD&A; GBPtoGBpunit 100; economic shares unchanged; active Q4 2027 targetabout GBp 47; analytical 42/60=70; protectedExcel 70. Review completion is separate from evidence completeness.